

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- 개념_*.pdf(슬라이드/치트시트)로 핵심 공식을 먼저 최종 점검한 뒤 풀이 시작

문항 목차 (총 12문항)

- [1] L05 · IPSRP 4.4.14 — 연속RV1
- [2] L05 · PSP 4.5.7 — 연속RV1
- [3] L06 · PSP 5.4.3 — 연속RV2
- [4] L06 · PSP 5.5.6 — 연속RV2
- [5] L07 · IPSRP 5.4.32 — Further1 변환·공분산
- [6] L07 · PSP 5.8.6 — Further1 변환·공분산
- [7] L07 · PSP 6.3.2 — Further1 변환·공분산
- [8] L08 · PSP 7.6.5 — Further2 상관·전분산
- [9] L09 · PSP 9.5.3 — 점근 CLT
- [10] 복습 L02 · PSP 2.2.5 — 조건부-Bayes
- [11] 복습 L03 · PSP 3.6.4 — 이산RV1
- [12] 복습 L04 · PSP 5.2.2 — 이산RV2

Problem 14

Let X be a random variable with the following CDF

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \leq x < \frac{1}{4} \\ x + \frac{1}{2} & \text{for } \frac{1}{4} \leq x < \frac{1}{2} \\ 1 & \text{for } x \geq \frac{1}{2} \end{cases}$$

- Find the generalized PDF of X , $f_X(x)$.
- Find EX using $f_X(x)$.
- Find $Var(X)$ using $f_X(x)$.

4.5.7 ● Y is an Erlang ($n = 2, \lambda = 2$) random variable.

- (a) What is $E[Y]$?
- (b) What is $\text{Var}[Y]$?
- (c) What is $P[0.5 \leq Y < 1.5]$?

5.4.3 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-(2x+3y)} & x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find $P[X > Y]$ and $P[X + Y \leq 1]$.
- (b) Find $P[\min(X, Y) \geq 1]$.
- (c) Find $P[\max(X, Y) \leq 1]$.

5.5.6■ Over the circle $X^2 + Y^2 \leq r^2$, random variables X and Y have the PDF

$$f_{X,Y}(x,y) = \begin{cases} 2|xy|/r^4 & x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the marginal PDF $f_X(x)$?
- (b) What is the marginal PDF $f_Y(y)$?

Problem 32

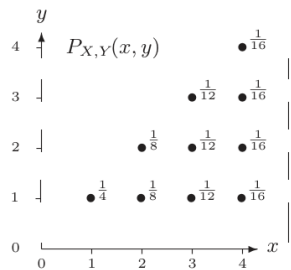
Let X and Y be two independent $N(0, 1)$ random variable and

$$Z = 11 - X + X^2 Y,$$

$$W = 3 - Y.$$

Find $\text{Cov}(Z, W)$.

5.8.6 ● The random variables X and Y have joint PMF



Find

- (a) The expected values $E[X]$ and $E[Y]$,
- (b) The variances $\text{Var}[X]$ and $\text{Var}[Y]$,
- (c) The correlation, $r_{X,Y} = E[XY]$,
- (d) The covariance, $\text{Cov}[X, Y]$,
- (e) The correlation coefficient, $\rho_{X,Y}$.

— 풀이 —

6.3.2■ In a 50 km cycling time trial, a rider's exact time T , measured in minutes, is the continuous uniform $(50, 60)$ random variable. However, a rider's recorded time R in seconds is obtained by rounding *up* T to next whole second. That is, if T is 50 minutes, 27.001 seconds, then $R = 3028$ seconds. On the other hand, if T is exactly 50 minutes 27 seconds, then $R = 3027$. What is the PMF of R ?

— 풀이 —

7.6.5 ♦ Use the iterated expectation for a proof of Theorem 5.19 without integrals.

— 풀이 —

9.5.3 ● The duration of a cellular telephone call is an exponential random variable with expected value 150 seconds. A subscriber has a calling plan that includes 300 minutes per month at a cost of \$30.00 plus \$0.40 for each minute that the total calling time exceeds 300 minutes. In a certain month, the subscriber has 120 cellular calls.

- (a) Use the central limit theorem to estimate the probability that the subscriber's bill is greater than \$36. (Assume that the durations of all phone calls are mutually independent and that the telephone company measures call duration exactly and charges accordingly, without rounding up fractional minutes.)
- (b) Suppose the telephone company does charge a full minute for each fractional minute used. Re-calculate your estimate of the probability that the bill is greater than \$36.

2.2.5 ● In a game of rummy, you are dealt a seven-card hand.

- (a) What is the probability $P[R_7]$ that your hand has only red cards?
- (b) What is the probability $P[F]$ that your hand has only face cards?
- (c) What is the probability $P[R_7F]$ that your hand has only red face cards? (The face cards are jack, queen, and king.)

3.6.4 ● At a discount brokerage, a stock purchase or sale worth less than \$10,000 incurs a brokerage fee of 1% of the value of the transaction. A transaction worth more than \$10,000 incurs a fee of \$100 plus 0.5% of the amount exceeding \$10,000. Note that for a fraction of a cent, the brokerage always charges the customer a full penny. You wish to buy 100 shares of a stock whose price D in dollars has PMF

$$P_D(d) = \begin{cases} 1/3 & d = 99.75, 100, 100.25, \\ 0 & \text{otherwise.} \end{cases}$$

What is the PMF of C , the cost of buying the stock (including the brokerage fee)?

— 풀이 —

5.2.2 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} c|x+y| & x = -2, 0, 2; \\ & y = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the value of the constant c ?

(b) What is $P[Y < X]$?

(c) What is $P[Y > X]$?

(d) What is $P[Y = X]$?

(e) What is $P[X < 1]$?